Preliminary Shielding Assessment - Refurbishment

A client-facing preliminary assessment for design development and shielding coordination. It identifies the adopted design basis, preliminary barrier requirements, and the information that must be confirmed before final shielding acceptance.

This report is not a certificate of shielding compliance.

1. Project details

Project name	Example Shielding Assessment
Room name	SPECT Room 1
Modality	SPECT
Project type	Refurbishment
Project note	Simplified preliminary shielding assessment.
Client	Example Imaging Pty Ltd
Site	123 Example Street, Sydney NSW
Assessor	Health Physics Pty Ltd
Date	2026-09-13
Report status	Preliminary
Revision	Rev 0

2. Assessment outcome

Overall preliminary outcome	**FAIL**
Barriers assessed	6
Barriers requiring review / redesign	3
Design decision	Do not rely on the current shielding arrangement for construction; redesign the failed barrier(s) and reassess.

3. Design basis

The preliminary result is valid only for the room geometry, source/workload and occupancy assumptions summarised below. These values should be confirmed by the client/design team before detailed design.

3.1 Room and geometry

Room	SPECT Room 1
Modality	SPECT
Room dimensions	5.00 m x 6.00 m
FFL to FCL	3.60 m
Source position	2.50 m from west | 2.60 m from south

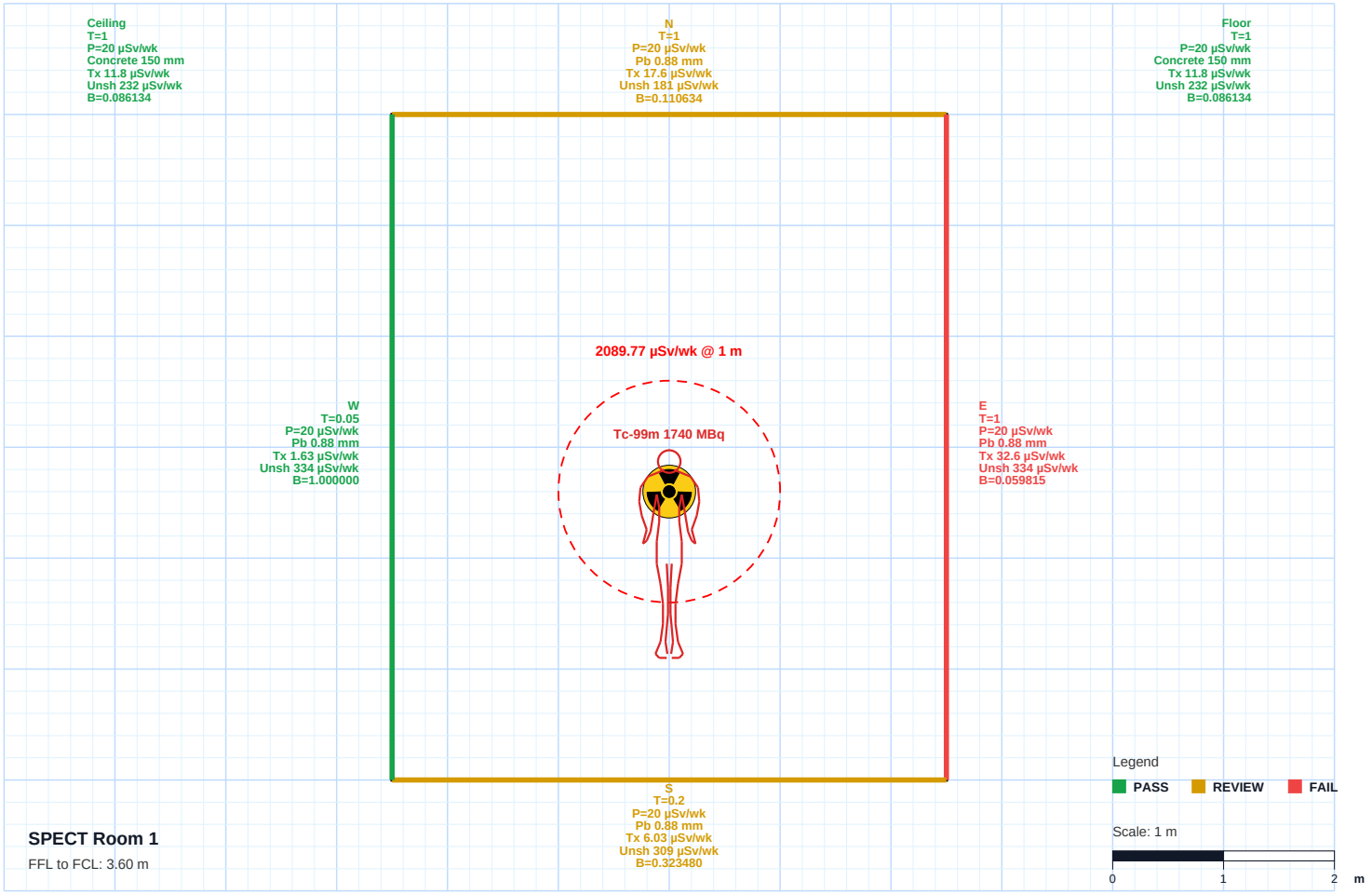
3.2 Source and workload assumptions

Parameter	Adopted preliminary input
Radionuclide	Tc-99m
Administered activity per patient	1740 MBq
Patient throughput	120 patients/week
Imaging duration	0.5 h/patient
Source position	2.50 m from west | 2.60 m from south

3.3 Design criteria and occupancy

Barrier	Adjacent area	Distance (m)	Occupancy T	P (uSv/wk)
N wall	Corridor	3.40	1.000	20.0
E wall	Corridor	2.50	1.000	20.0
S wall	Corridor	2.60	0.200	20.0
W wall	Corridor	2.50	0.050	20.0
Floor	Clinical Area	3.00	1.000	20.0
Ceiling	Clinical Area	3.00	1.000	20.0

4. Barrier plan



Barrier labels show the entered construction and the calculated preliminary result. The architectural drawing, openings and service penetrations must be coordinated separately before final acceptance.

5. Barrier schedule

The schedule below reports the canonical calculation results for the assessed barriers.

Barrier	Actual (mm)	Role	Distance (m)	T	P (uSv/wk)	Unshielded (uSv/wk)	Transmitted (uSv/wk)	B	Required (mm)	Solver	Status
N wall	0.88	Secondary wall	3.40	1.000	20.000	180.776	17.630	0.110634	0.83	spect_tc99m_practical_archer	REVIEW
E wall	0.88	Secondary wall	2.50	1.000	20.000	334.364	32.609	0.059815	1.07	spect_tc99m_practical_archer	FAIL
S wall	0.88	Secondary wall	2.60	0.200	20.000	309.138	6.030	0.323480	0.41	spect_tc99m_practical_archer	REVIEW
W wall	0.88	Secondary wall	2.50	0.050	20.000	334.364	1.630	1.000000	0.00	spect_tc99m_practical_archer	PASS
Floor	150.00	Floor	3.00	1.000	20.000	232.197	11.770	0.086134	128.40	spect_tc99m_practical_archer	PASS
Ceiling	150.00	Ceiling	3.00	1.000	20.000	232.197	11.770	0.086134	128.40	spect_tc99m_practical_archer	PASS

6. Design actions and verification

This page identifies the items that should be closed out before the preliminary assessment is relied upon for construction or final shielding acceptance.

6.1 Project coordination actions

Action	Suggested responsibility	Required before
Verify existing wall, floor and ceiling construction/thickness against the entered shielding values.	Client / builder; physicist review	Final assessment
Confirm source workload, occupancy classification and installed barrier thickness.	Client / design team / physicist	Design freeze
Increase shielding or revise source / occupancy assumptions.	Client / design team / physicist	Design freeze
Confirm source workload, occupancy classification and installed barrier thickness.	Client / design team / physicist	Design freeze
Confirm the current architectural drawing number/revision, room orientation and dimensions.	Architect / client	Detailed assessment
Confirm the use and occupancy of areas beyond each wall, floor and ceiling.	Client / architect	Detailed assessment
Provide door, viewing panel, duct, conduit and service-penetration details so shielding continuity can be assessed.	Architect / builder	Construction documentation
Confirm radionuclide/scanner workload and operating assumptions with the intended operator.	Imaging operator / client	Detailed assessment

6.2 Changes that trigger reassessment

- Any increase in administered activity, patient throughput or time spent in the assessed source position.
- Any change to room dimensions, source position, adjacent-area use or occupancy.
- Any change to barrier construction, doors, glazing, penetrations or shielding continuity.
- Addition of radiation-producing equipment or a source not represented in this assessment.

7. Conclusion and next stage

Overall preliminary outcome	**FAIL**
Barrier result	3 PASS | 2 REVIEW | 1 FAIL
Client decision	Do not rely on the current shielding arrangement for construction; redesign the failed barrier(s) and reassess.

7.1 Preliminary conclusion

The preliminary assessment contains 1 barrier failure(s). The affected barriers require redesign and reassessment before the shielding arrangement is accepted for construction.

7.2 Recommended next stage

For a refurbishment, the next stage should verify existing barrier construction, thickness, openings, penetrations and adjacent-area use against coordinated architectural information before issuing a detailed/final shielding assessment.

7.3 Information normally required for the detailed/final assessment

- Current coordinated architectural drawings showing room dimensions, wall construction, doors, glazing and service penetrations.
- Evidence or site verification of existing wall, floor and ceiling construction and shielding thickness/equivalence.
- Confirmed use and occupancy of all adjacent areas, including above and below the room.
- Confirmed clinical workload/source assumptions and any additional radiation-producing equipment.

8. Preliminary assessment limitations

- This document is a preliminary shielding assessment for design development and shielding review; it is not a final shielding compliance certificate.
- Final shielding acceptance requires professional review of architectural drawings, construction details, source and workload assumptions, occupancy assumptions, and installed materials.
- Calculated results depend on the entered room geometry, source position, operating assumptions, design goals, occupancy factors, and barrier construction.
- SPECT results depend on the selected radionuclide, administered activity, patient throughput and timing assumptions, source location, and occupancy model.
- Any additional radiation-producing equipment or operational source not represented in this assessment requires separate evaluation before final acceptance.